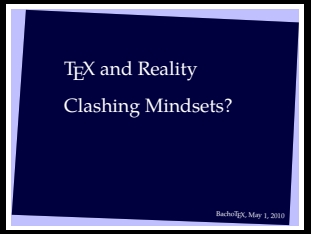
# Presentations

This is a selection of three sets of presentations at meetings and a bunch of examples for some styles that come with ConTeXt. In MkIV some of the older styles have been dropped. There are more presentations, starting mid 1990's: some are lost, others make no sense showing (or have articles instead) and I have no time anyway to collect all of them.

We show the first four pages but you can find the complete files in the ConTeXt distribution. This file is generated automatically so it's also a snapshot. Maybe some day I find the time to add some more.

There are more files in the source tree because it doesn't make sense to provide all pdf files. Some files are to be generated with LuaTeX because they discuss something (MkIV) that in the meantime has been superseded. Keep in mind that these talk are snapshots and might have presenting the state at that time, some experiments, or just have been teasers.

## bachotex-2010-clash





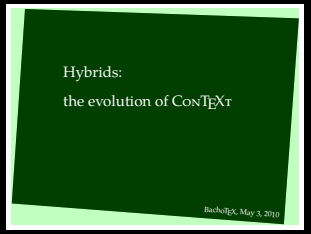






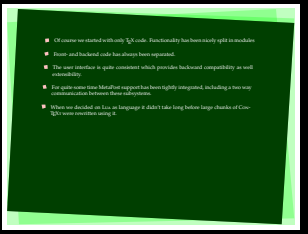


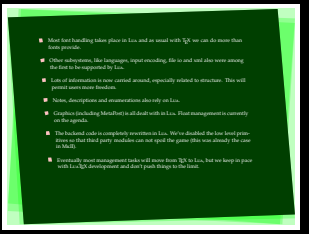
## bachotex-2010-move





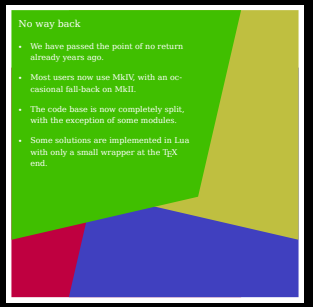





## bachotex-2011-cld-and-mkvi





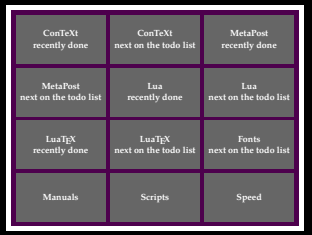


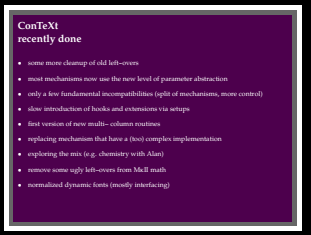



BachoTEX

# bachotex-2013-bits



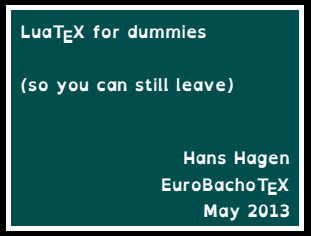



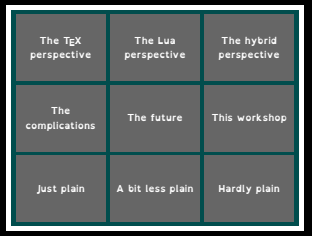



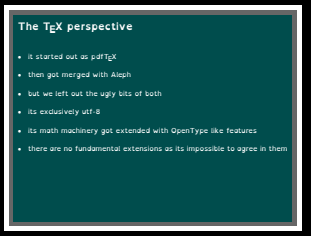


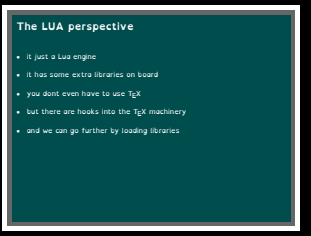


# bachotex-2013-luatex

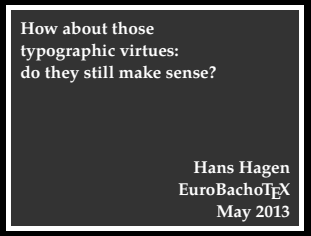


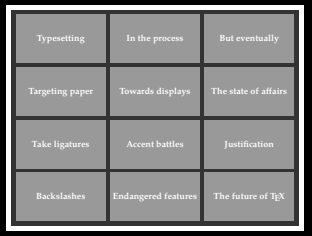



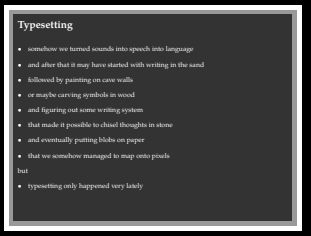



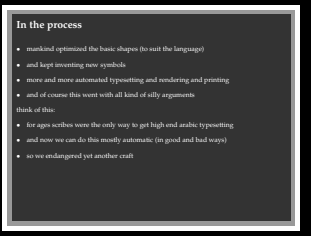




# bachotex-2013-sense

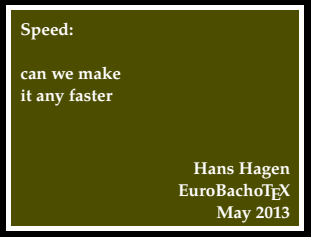


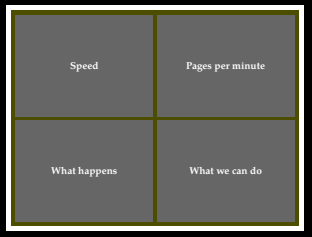



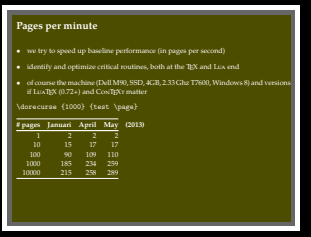
## bachotex-2013-speed



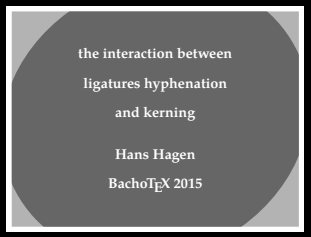



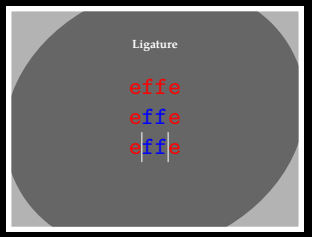


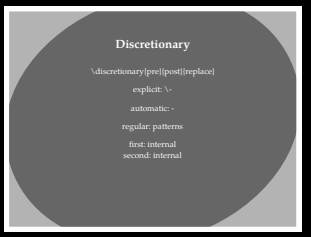


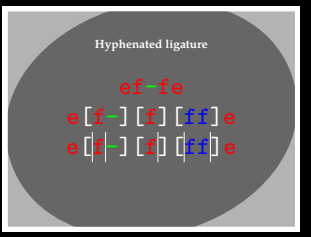



## bachotex-2015-ligatures

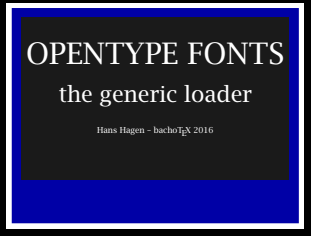


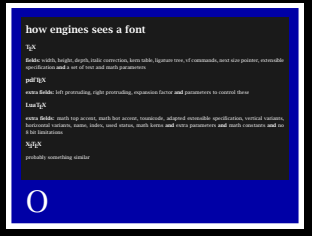



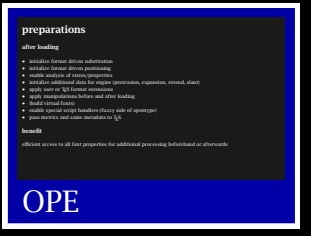

## bachotex-2016-opentype










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# bachotex-2016-toolbox

**Grandpa's toolbox**

(making closets)

**Dad's hobby**

(cleaning closets)

**Uncle's friends**

(talking closets)

**bachoTEX 2016**

– +

- there's closets and closets
- take the ones you put stuff in ... e.g. that you buy at ikea
- you have to assemble them so you get out your toolbox
- this kid sits next to you wondering what that tool is
- what is natural for you to use might not be so for them
- but grandpa likes to carry over his knowledge and experiences

– **closets** +

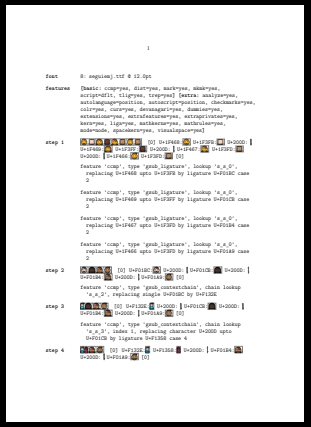
- but ... kids get their information from the internet, not from you
- (they watch discovery channel or national geographic and know a lot)
- or they look at vloggers (no bloggers) trying to learn something
- and they keep moving on ... and on
- do gp's live long tools really make sense to them?

– **interest** +

- grandpa also has a vlogger, he is called knuth
- gp is as locked into tex as the kids are into apps
- he's a do-it-him-self kind of person
- will his grandkids love watching him crafting
- and hear his stories about meetings and journals

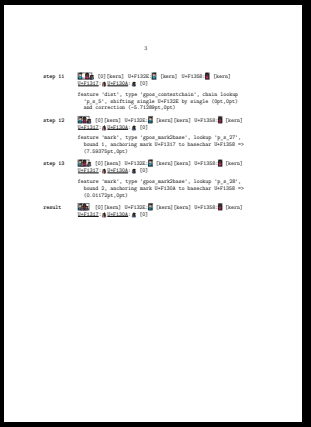
– **grandpas** +

# bachotex-2017-emoji-demo










# bachotex-2017-emoji

## Picture Fonts

**welcome to a (beautiful) mess**

**Hans Hagen**
**BachoTUG 2017**

## A Summary

- **the macro package's view:** just a font like any other but it needs to configure some extra color related properties
- **the engine's view:** depending on the technology a normal font that needs a bit special treatment or needs to be dealt with as collection of graphics
- **the viewer's view:** regular outline glyphs or images tagged as kind of characters so that their unicode representation can be cut and paste
- **the user's view:** more pictures than glyphs although some people one can communicate using them

So, in practice, for most TEX users it's probably not a high priority font but more a fun one.

## Technologies

As each vendor came up with something, we have to deal with a all kinds of formats. And or course, as eagerness pushes things on the market before it's perfect we now have to deal with all of them.

- **overlapping glyphs:** this technique uses the `colr` and `cpal` tables and is actually a quite clean technology, you can combine in different ways
- **svg graphics:** this technique uses the `svg` table that contains a svg vector image
- **bitmap graphics:** this technique uses for instance `sbix` tables that can have various graphic images

The first two are already supported in the CONTEXT font loader and processor for a while, the last one was added recently.

Only the overlapping method is useable for the tens of thousands of skin tone combinations of families, (kissing) couples, and professions.

## Preparation

For now one has to enable the feature:

```
\definefontfeature[overlay][default][ccmp=yes,colr=yes,dist=yes]
\definefontfeature[svg]    [default][svg=yes]
\definefontfeature[bitmap] [default][sbix=yes]

\definefontfeature [colored] [default]
  [ccmp=yes,dist=yes,
   colr=yes,svg=yes,sbix=yes]
```

Defining a font is not different from others

```
\definefont[MyEmojiFont] [seguiemj*overlay]
\definefontsynonym[emoji][seguiemj*overlay]
```

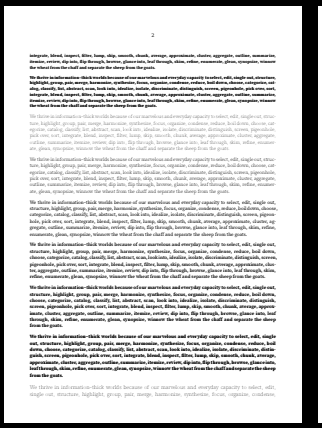
As is using:

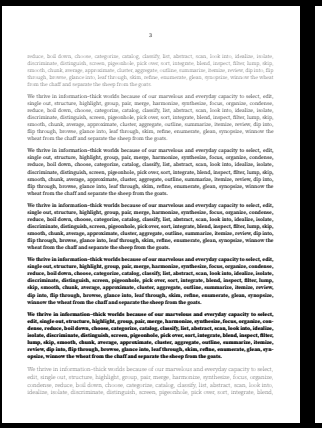
```
{\MyEmojiFont\resolvedemoji{woman}}
\emoji{woman}
```

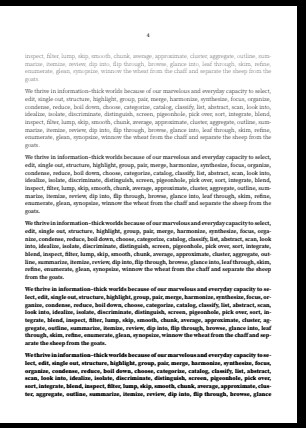
BachoTEX

# bachotex-2017-variablefonts-demo









# bachotex-2017-variablefonts

## Variable Fonts

**we're ready for them**

**Hans Hagen**
**BachoTUG 2017**

## A Summary

- **the macro package's view:** just a font but with many possible variations in shapes (width, weight, slope, etc) and therefore a bit more complex user interface
- **the engine's view:** an abstraction not different from other fonts but that needs a special treatment in the backend
- **the viewer's view:** a font to be displayed like any other with outlines in cff of ttf format
- **the user's view:** an opentype font with possibly surprising shapes of which you need to know a bit more than usual if you want to profit from it

So, in practice, for most TEX users it's just a font that has to be supported by TEX and friends.

## Starting point

- The OpenType 1.8 specification at the MicroSoft website defined the extra tables and explains bits and pieces.
- There a few fonts that have relevant tables (not all) and implement variants as well as features.
- There are some posts on the internet that show a bit about axis and other things that go on in these fonts.
- Luckily we have ways (in CONTEXT) to explore what goes on in these fonts and how they could look.
- Condition: no tricks, no fuzzy heuristics, just the specification should be enough.

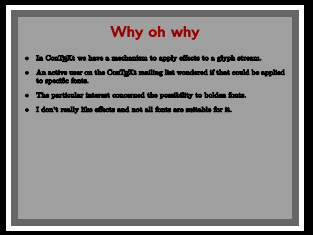
## Implementation steps

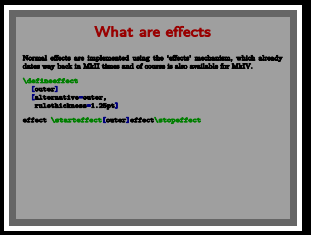
- First try to render variants in order to see what we're dealing with. This was not too hard (starting with cff) because we have already virtual font support.
- Next try to load the relevant tables and figure out what these deltas and such really mean and how axis and regions and ... have to be applied.
- Try to make it all work on a real piece of text, so not only shapes but also features and dimensions.
- Finally make sure that the font can get embedded as a normal font and not as inline (tagged) graphic.
- Also, try to generalize the helpers and methods in such ways that we can experiment with additional tricks (after all, TEX is about control).
- Todo: once there are more fonts (with the right data tables), check the code with the specification.

BachoTEX

## bachotex-2018-fonteffects



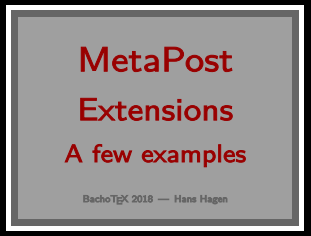



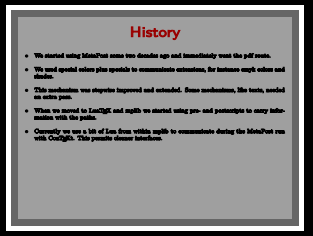







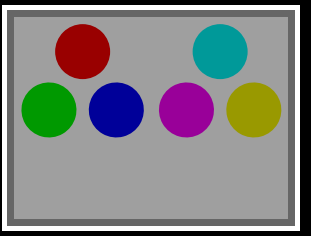
## bachotex-2018-mp



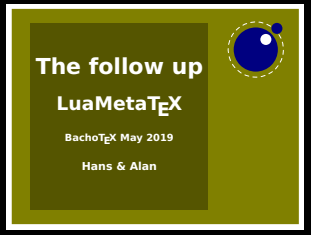







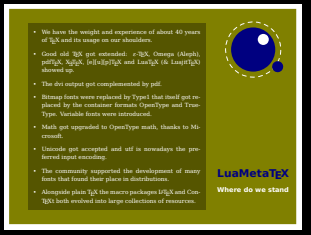


## bachotex-2019-followingup





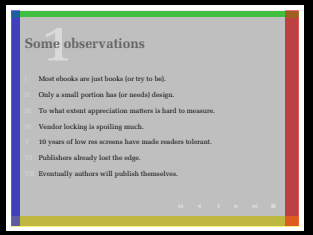


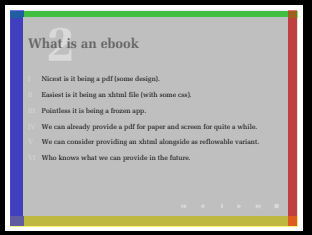


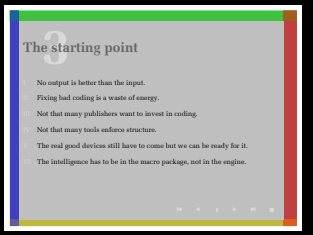



context-2011-ebook-export

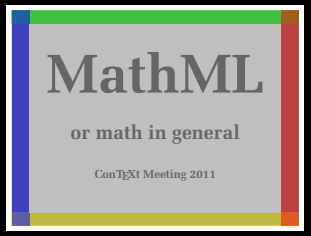


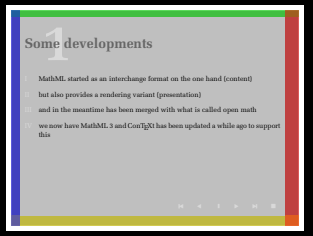



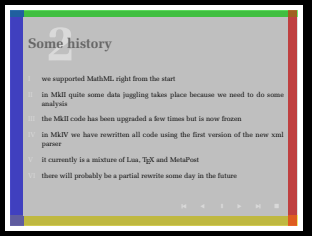



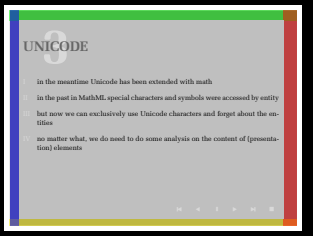




context-2011-mathml-update



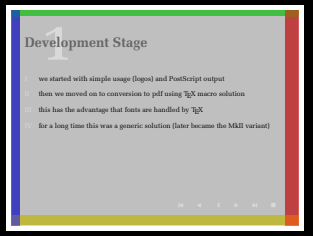



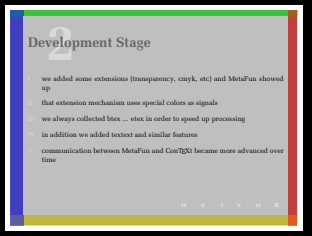



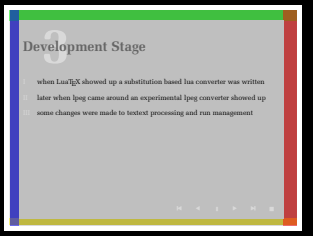




context-2011-metapost-how-we-adapt



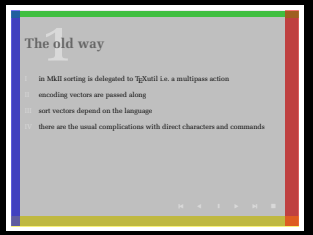



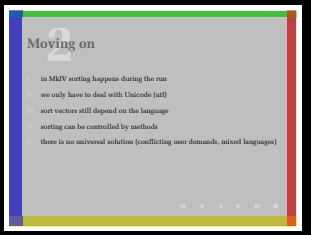



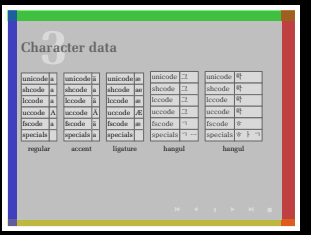



## context-2011-sorting-registers




## context-2012-after-the-cleanup





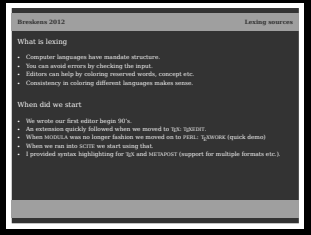


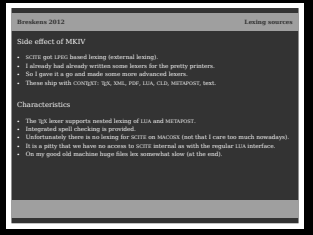






## context-2012-lexing-sources



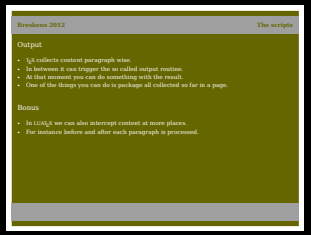



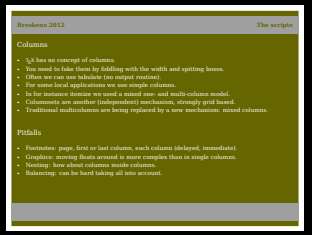




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## context-2012-mixed-columns

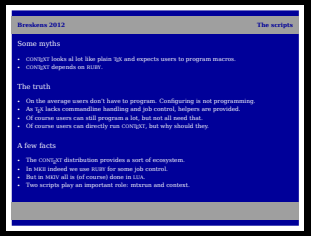


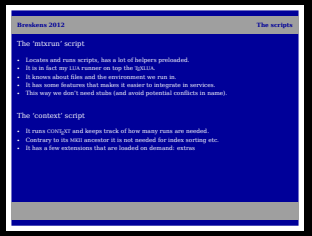



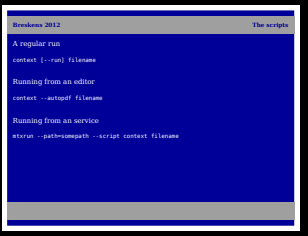


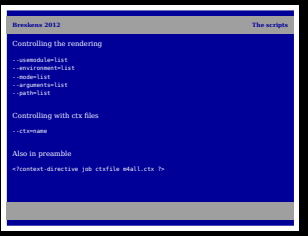


## context-2012-the-script



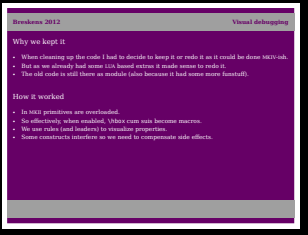




## context-2012-visual-debugging





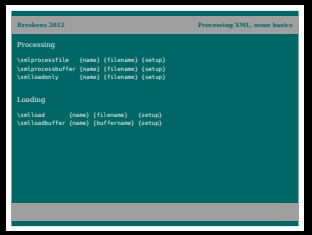


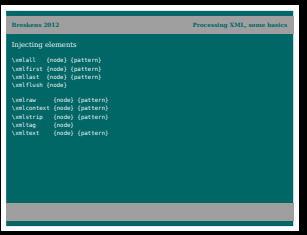


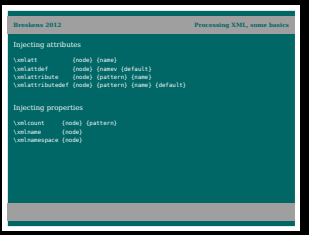



# context-2012-xml-news



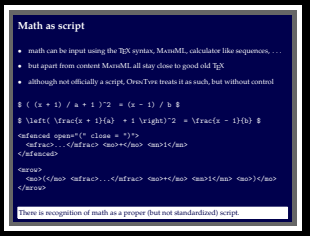



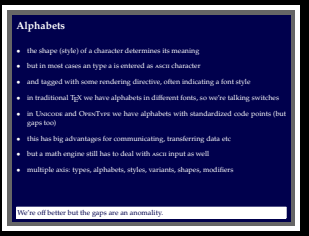

# context-2013-math





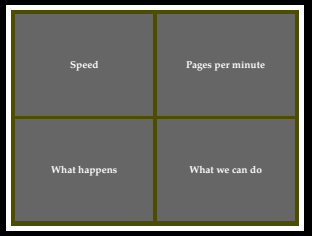


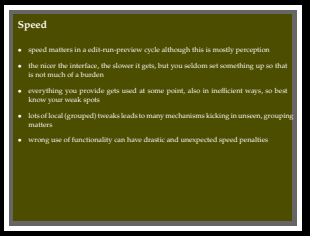


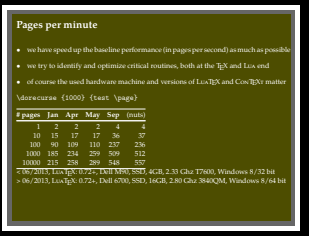




# context-2013-speed



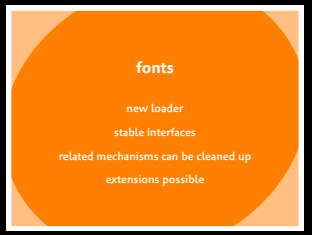



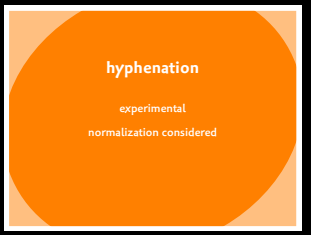



## context-2015-status



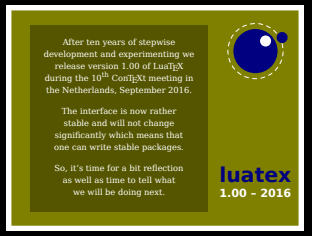



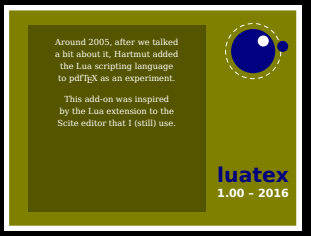


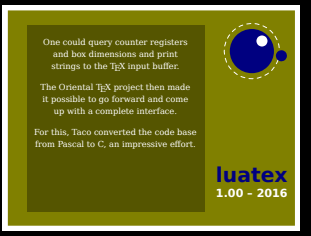


## context-2016-luatex

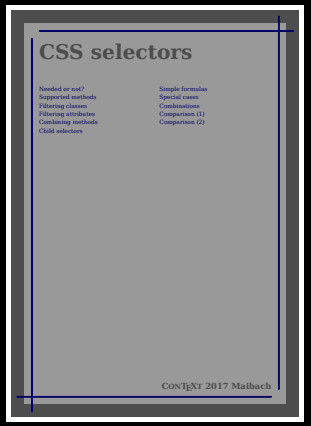


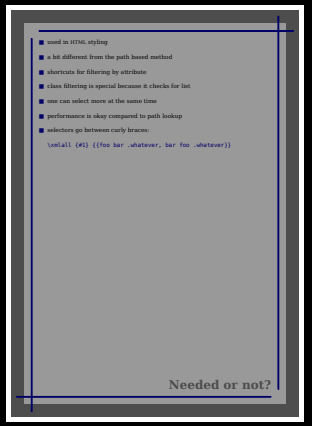



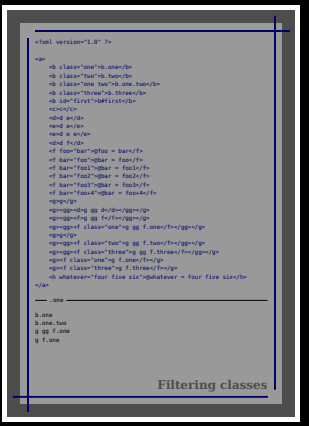
context-2017-css-selectors










context-2017-features-chaintest




...

context-2017-features-kerntest




...

context-2017-features-pairtest




...

context-2017-features-singletest



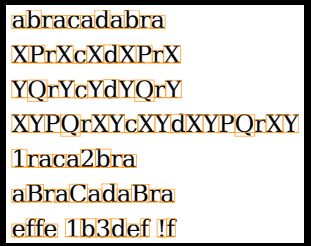

...

context-2017-features-spacetest




...

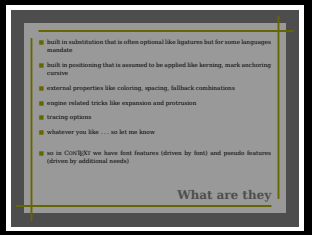
context-2017-features-substitutiontest

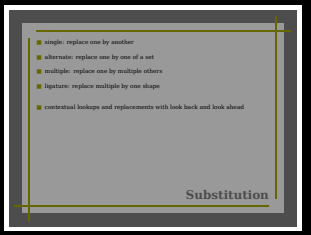


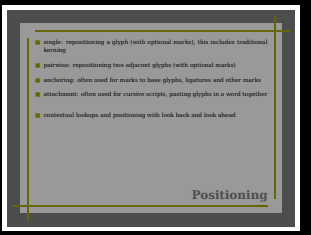

...

## context-2017-features



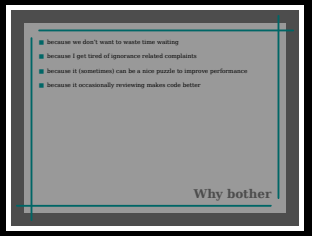



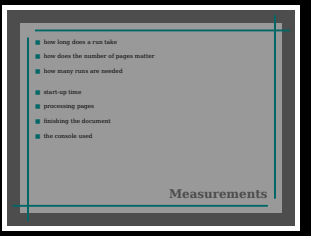

## context-2017-performance



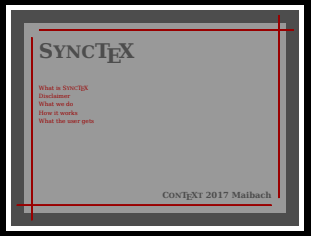







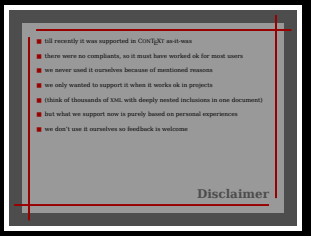



## context-2017-synctex



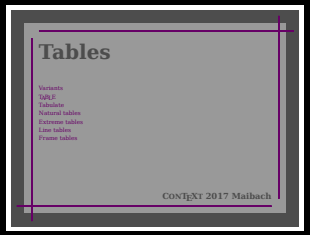


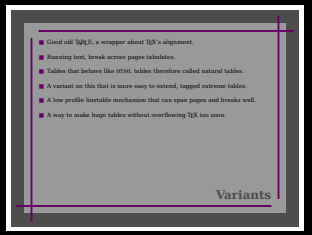


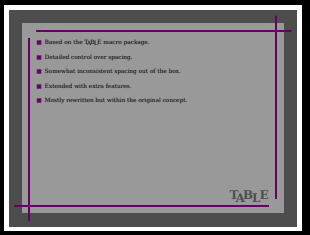


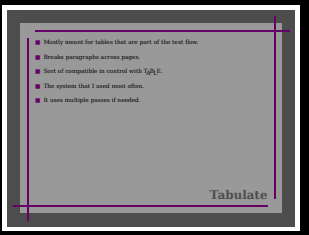



## context-2017-tables

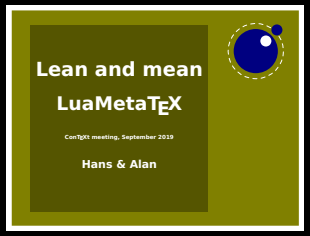


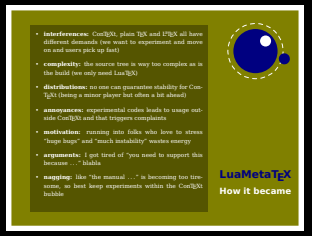



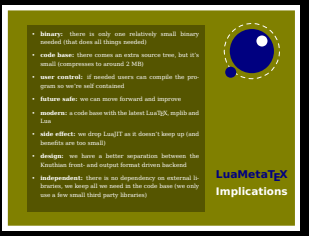

## context-2019-lmtx



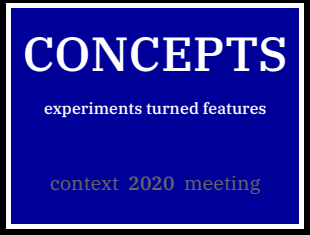



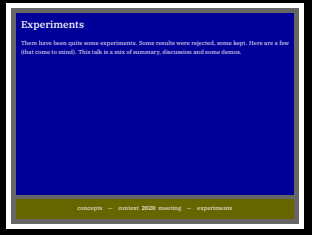


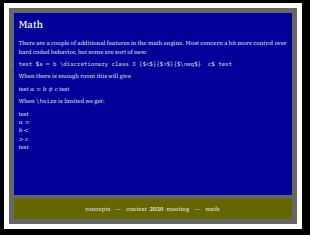





## context-2020-concepts



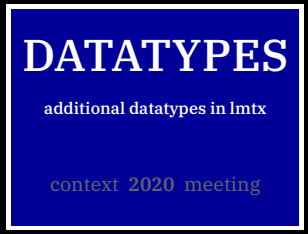



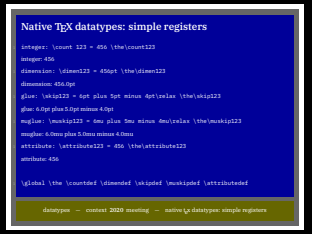


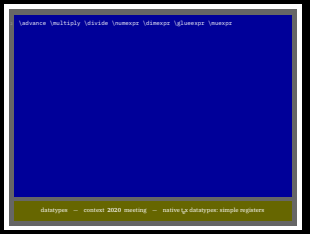


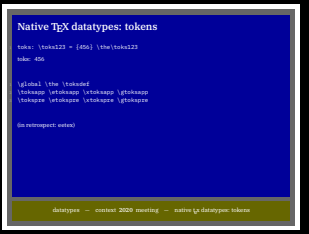
ConTEXt Meeting

## context-2020-datatypes



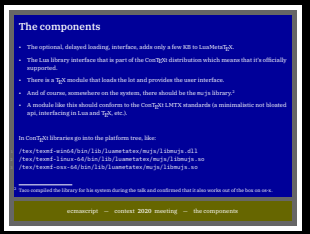



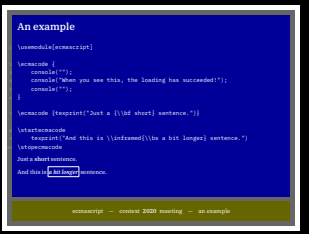

## context-2020-ecmascript









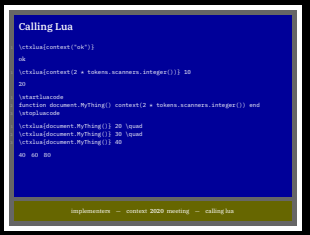




## context-2020-implementers













ConTEXt Meeting

## context-2020-luametatex

# LUAMETATEX

## where do we stand

context 2020 meeting









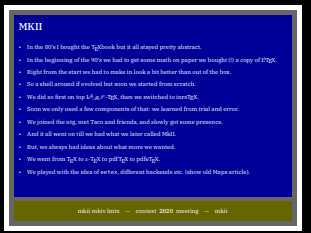

## context-2020-mkii-mkiv-mkxl

# MKII MKIV LMTX

## where does it end

context 2020 meeting







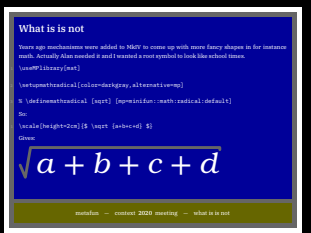
## context-2020-mp

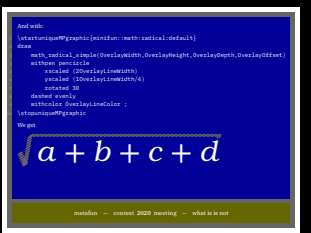
# METAFUN

## simple fonts

context 2020 meeting



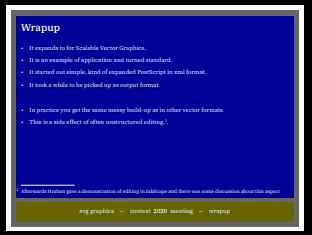




ConTEXt Meeting

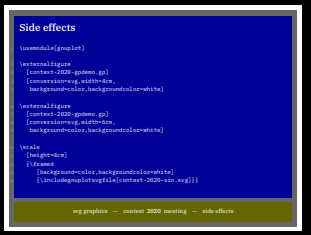
## context-2020-svg










## context-2020-tokens





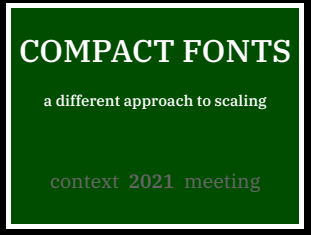






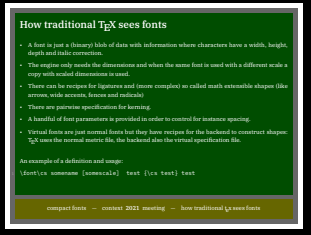


## context-2021-compactfonts













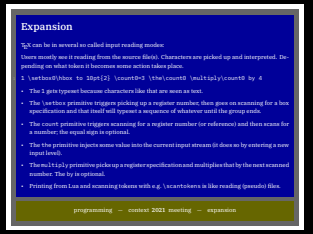
ConTEXt Meeting

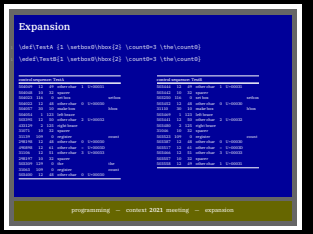
context-2021-localcontrol

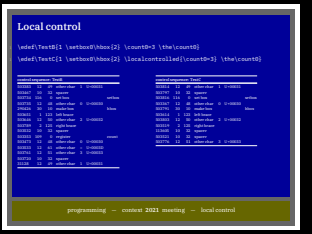
# PROGRAMMING

## local control

context 2021 meeting



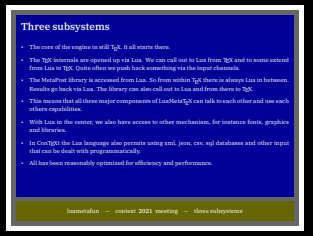




context-2021-luametafun

# LUAMETAFUN

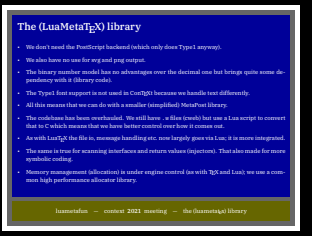
## the new interfaces

context 2021 meeting







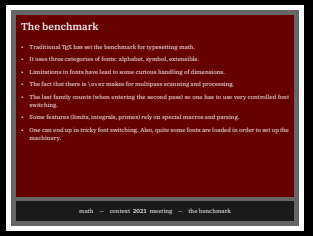



context-2021-math

# MATH

## complete control

context 2021 meeting









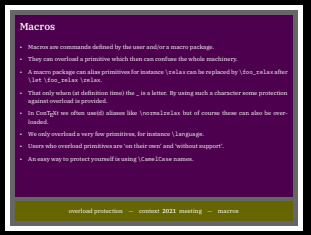

ConTEXt Meeting

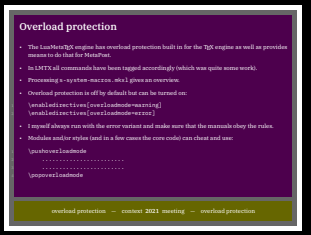
# context-2021-overloadprotection









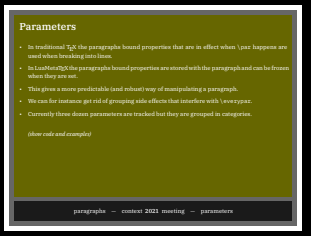

# context-2021-paragraphs





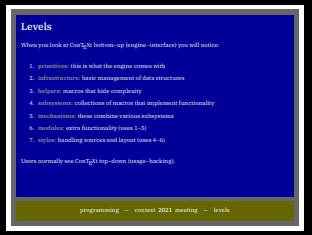


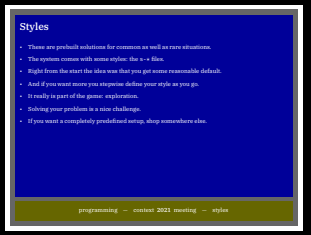


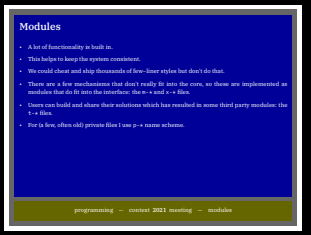




# context-2021-programming



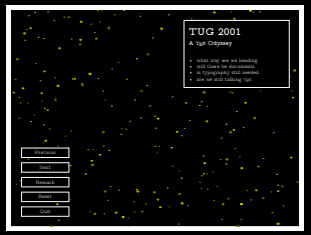



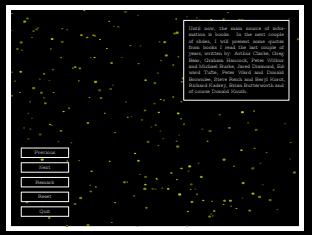



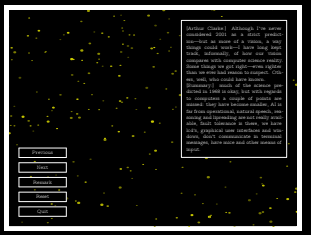



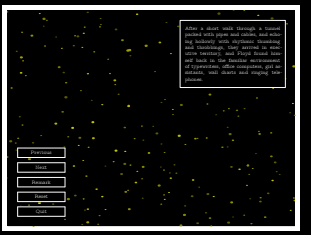

ConTEXt Meeting

# tug-2001-ideas



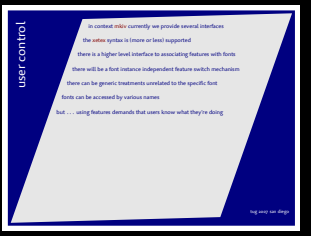

# tug-2007-fonts



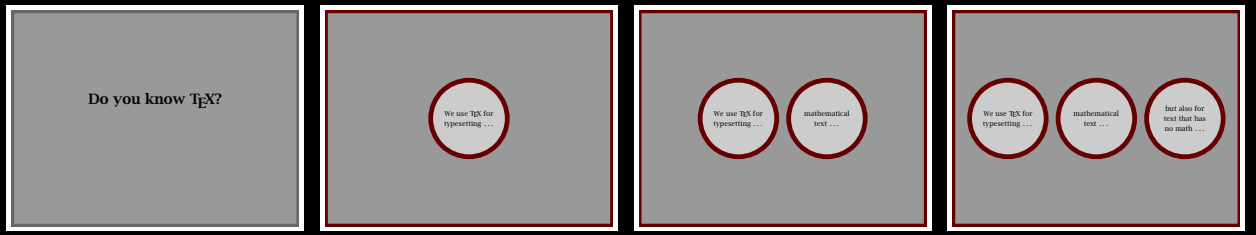




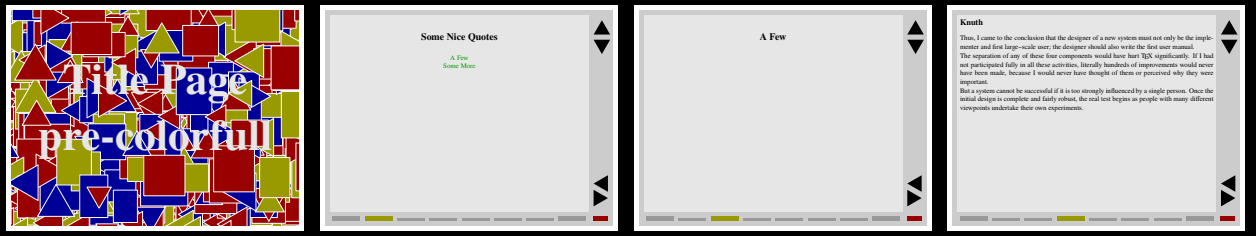


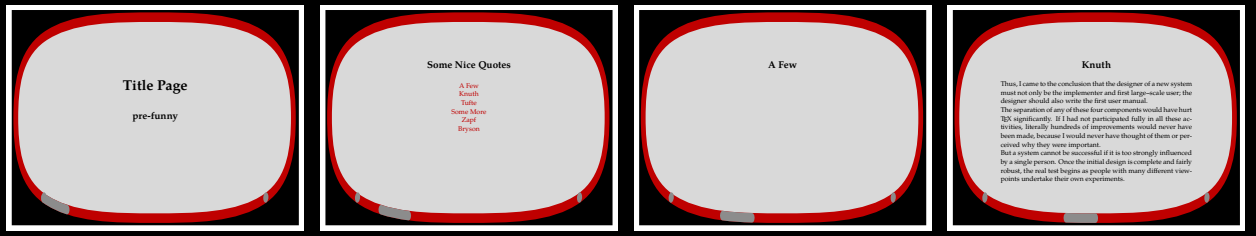
## present-balls-001




## present-colorful-001




## present-funny-001



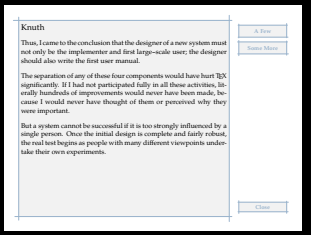

Examples

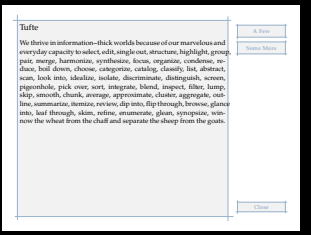
## present-fuzzy-001









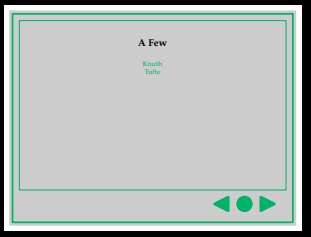



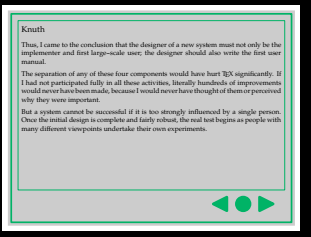

## present-green-001



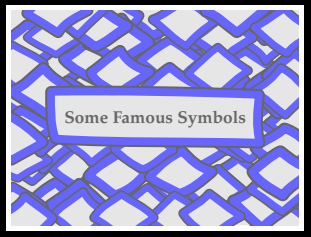


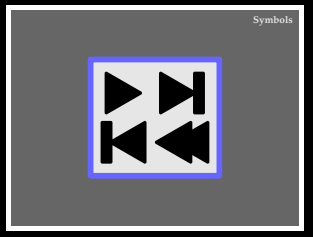


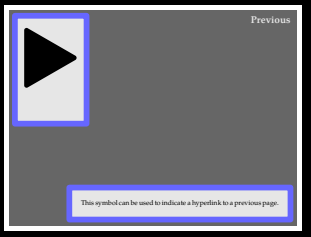


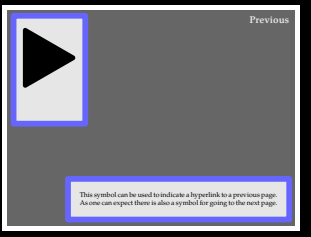




## present-grow-001



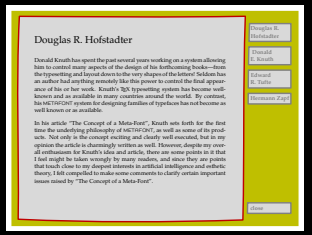



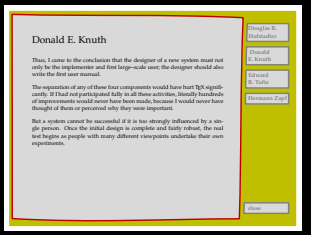



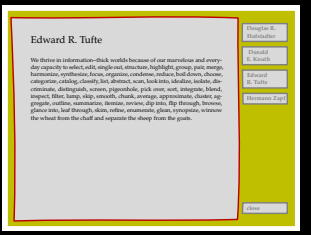

Examples

## present-organic-001

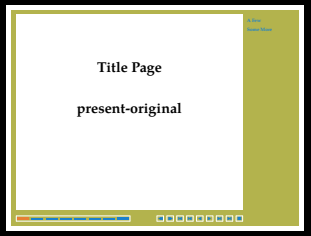


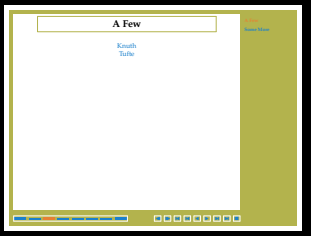



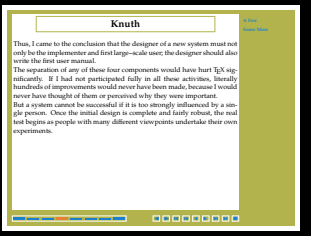

## present-original-001










## present-punk-001







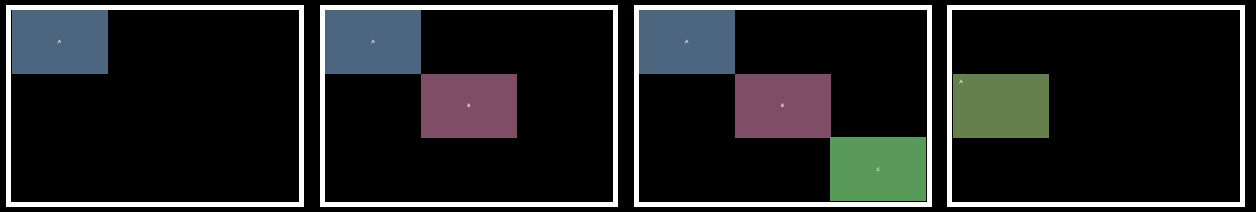






Examples

present-random-001




present-shaded-001



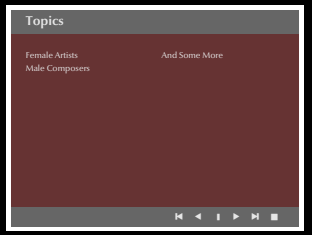

present-split-001

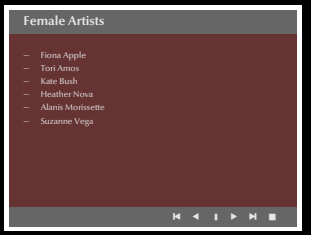



Examples

## present-stepper-001



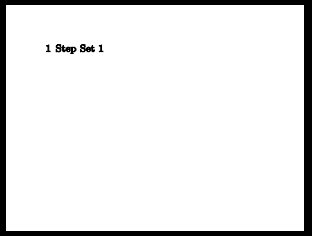



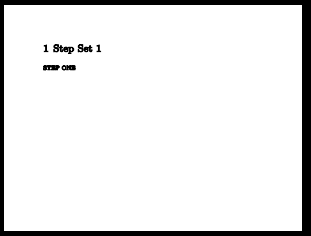


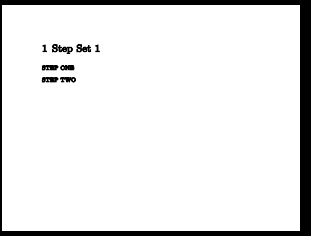


## present-steps-001



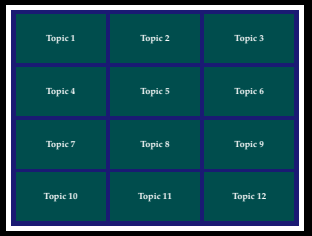



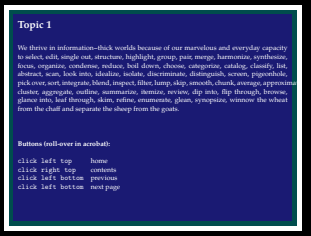



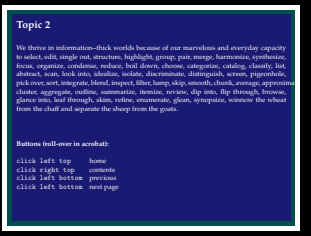




## present-tiles-001



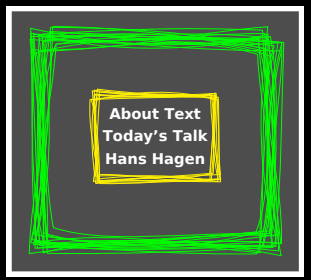



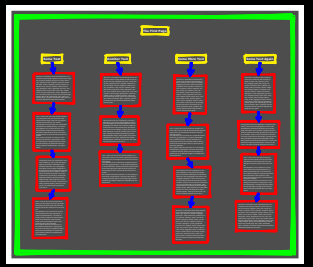



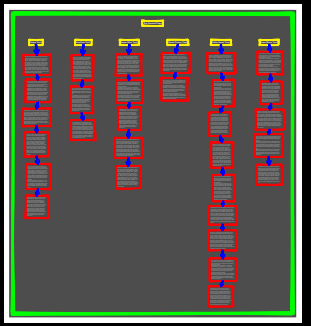



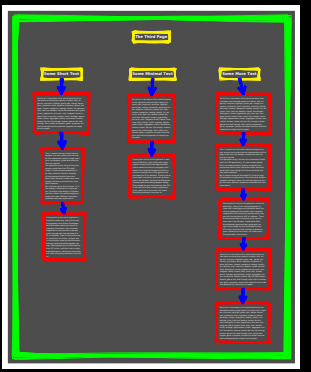

Examples

present-weird-001





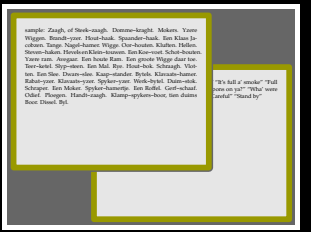


present-windows-001







Examples